



Productivity Improvement of The Lighthouse Wind Farm

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Bootcamp Unicorn Academy 9.0
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About Me



I am a Chemical Engineer specialized in continuous improvement, data analytics, process optimization and business intelligence solutions.

My work focuses on transforming complex data into actionable insights through statistical analysis, process improvement methodologies and data visualization techniques to enhance operational performance and support strategic decisions.

AEROWIND

Aerowind contacted me to conduct a comprehensive operational performance analysis of the 'Lighthouse' Wind Farm, strategically located in the coastal region of Buenos Aires Province, an area known for its strong, stable and high-frequency wind regime, making it highly suitable for wind energy generation.



Strategically located in the coastal region of Buenos Aires, Argentina, known for its intense and stable wind regime.



Installed Turbines

2023



5

Vestas V100

2024



2

Vestas V110

2025



3

GE -2.5 116



10 *wind turbines*



21,9 MW *installed capacity*



48 GWh *annual energy*

22

AEROWIND lacks an integrated system to objectively measure its operational performance. Data is currently fragmented across multiple sources, leading to visibility gaps and inconsistent decision criteria across the organization.

CUSTOMER NEEDS

- Define and calculate the farm's main KPIs.
- Design an interactive dashboard for real-time performance monitoring to support strategic decisions.
- Provide a reliable, data-driven tool to replace manual reports and enable more precise and efficient management.
- Answer business questions and increase energy efficiency.

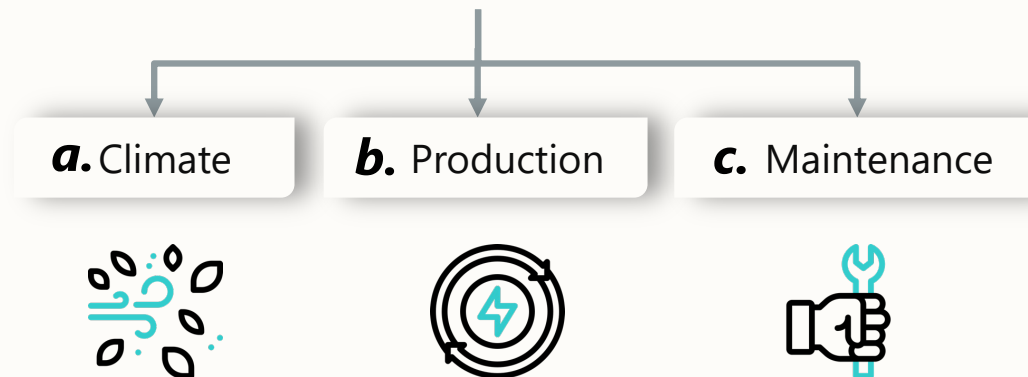
|| 2. WIND FARM CHARACTERISTICS



Objective

The main objective is to evaluate the wind farm's operational efficiency and historical performance, identifying production deviations and analyzing the impact of weather, operational and maintenance factors.

3 essential components were analyzed



a

Due to wind variability, historical data analysis is essential to identify performance trends and seasonal patterns.

b

Production fluctuations allow identification of energy losses and understanding of real performance.

c

Proper maintenance planning impacts turbine availability and energy generation.



Tools Used

A



Chat GPT

Database generated with artificial intelligence.

B



MySQL

Data validation and cleaning.

C



Power BI

Data visualization.

A

6 databases were generated:

1)  production

2)  maintenance

3)  climate

4)  turbines

5)  price

6)  theoretical_curves

FACT TABLES

DIMENSION TABLES

B

Cleaning & queries in MySQL.

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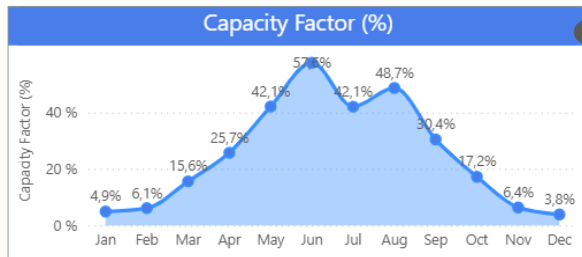
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Dashboard for data visualization.







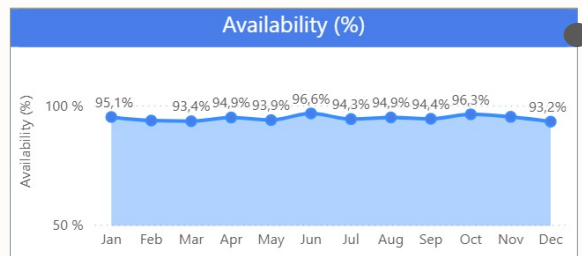


Capacity Factor (%)

Turbine energy efficiency.

$$CF = \frac{\text{Real energy production (kWh)}}{\text{Maximum possible energy production (kWh)}}$$

TARGET
25%

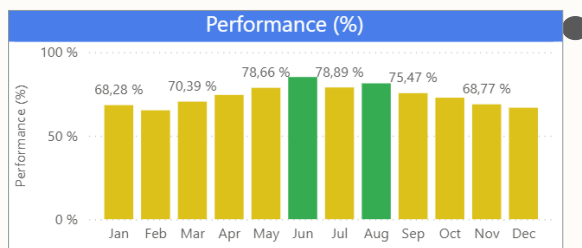


Availability (%)

Percentage of time the turbine remained operational.

$$\text{Availability} = \text{Operative Hours} / \text{Total Hours}$$

TARGET
90%

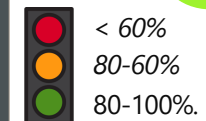


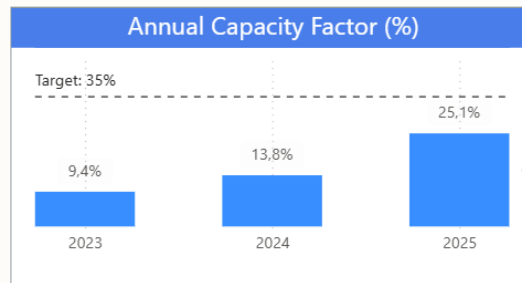
Performance (%)

Main Metric. Combines the two above.

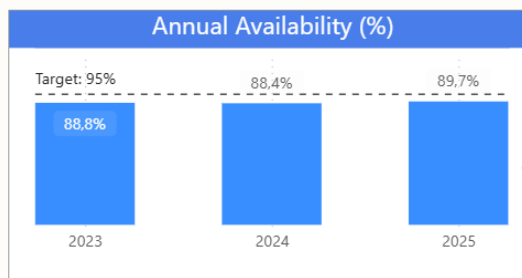
$$\text{Performance} = 0.65 \cdot D + 0.30 \cdot F + 0.05 \cdot P$$

TARGET
80%

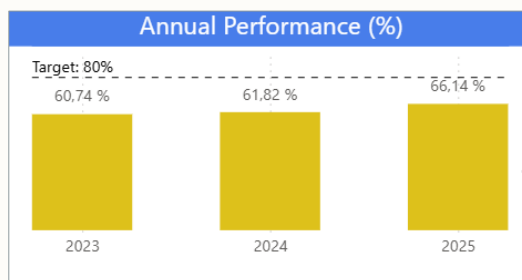




The **Capacity Factor** increases from 9,4% to 25,1%, driven mainly by the incorporation of new technologies, not by changes in wind resource. *It approaches the target but can still improve significantly.*



Availability remains at high levels close to 90% year after year. *A more challenging target of 95% has been set.*



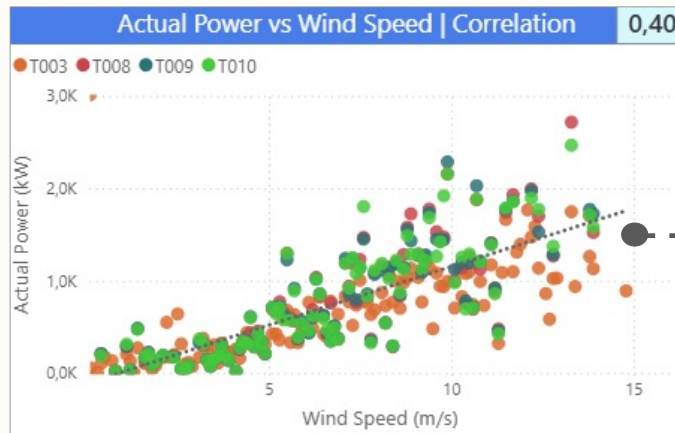
The **Performance** remains stable at acceptable values close to 65%. *The goal is to bring it to optimal values of 80%.*

|| 6. EXPLORATORY ANALYSIS





Turbine Performance Identification



Turbine T003 shows real power values below the newer units (**T008, T009, T010**) at high wind speeds, where generation potential is greatest.

This gap indicates that while T003 performs comparably at low wind speeds, it fails to capture the full generation potential available at stronger wind conditions, resulting in below-expected energy output relative to the modern fleet.

Conclusion: T003 underperforms compared to newer turbine models.



This chart reinforces the diagnosis: T003 stands as an outlier, combining one of the highest maintenance costs with below average availability.

Conclusion: T003 has the highest maintenance costs and lowest availability in the farm.

In contrast, turbines installed last year (**GE 2.5-116**) show optimal availability and cost levels.

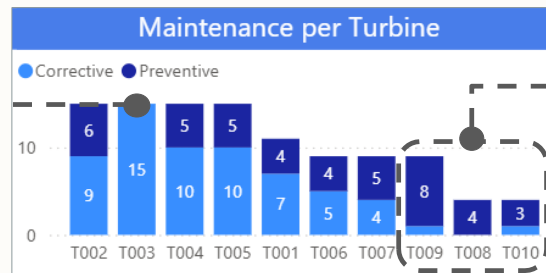
Conclusion: T008/T009/T010 are consolidated as benchmark turbines of the wind farm.





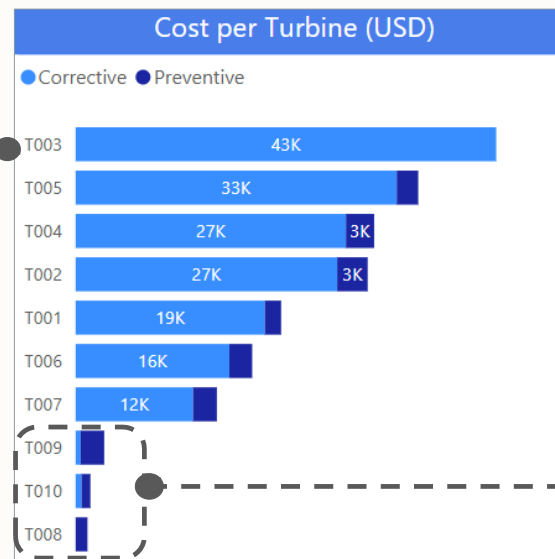
Operational Reliability

T003 records the highest number of corrective maintenance events in the fleet, with no preventive interventions recorded.



T008, T009 and T010 show minimal maintenance activity and no corrective interventions, with preventive maintenance predominating, indicating strong operational reliability.

Highest costs: **T003**



This is explained by the lower cost of preventive maintenance compared to corrective maintenance:

- AVG preventive: \$467 USD
- AVG corrective: \$2872 USD



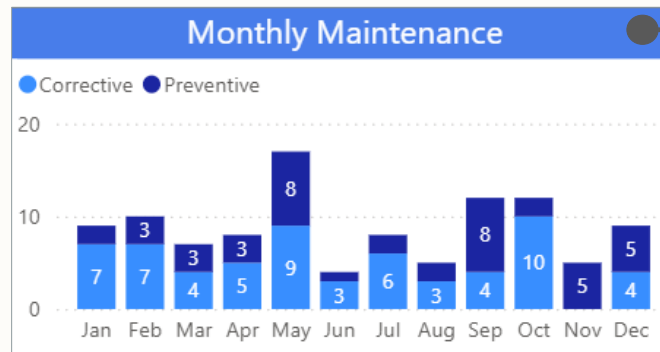
WHY?

Lowest costs: **T008, T009 y T010**



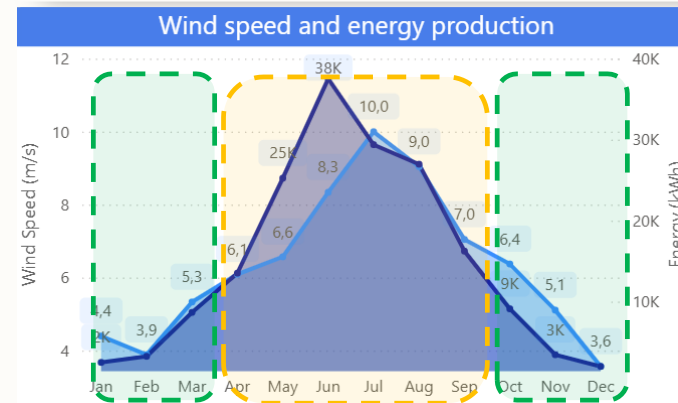
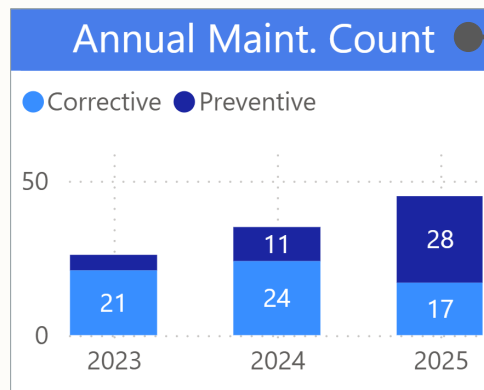


Operational Reliability



Maintenance activities are not strategically scheduled, creating **opportunities for operational optimization**.

Preventive maintenance should be concentrated during low-production periods to minimize interventions during peak generation hours.



- High-wind months with greatest energy generated
- Low-production months, optimal for performing maintenance.

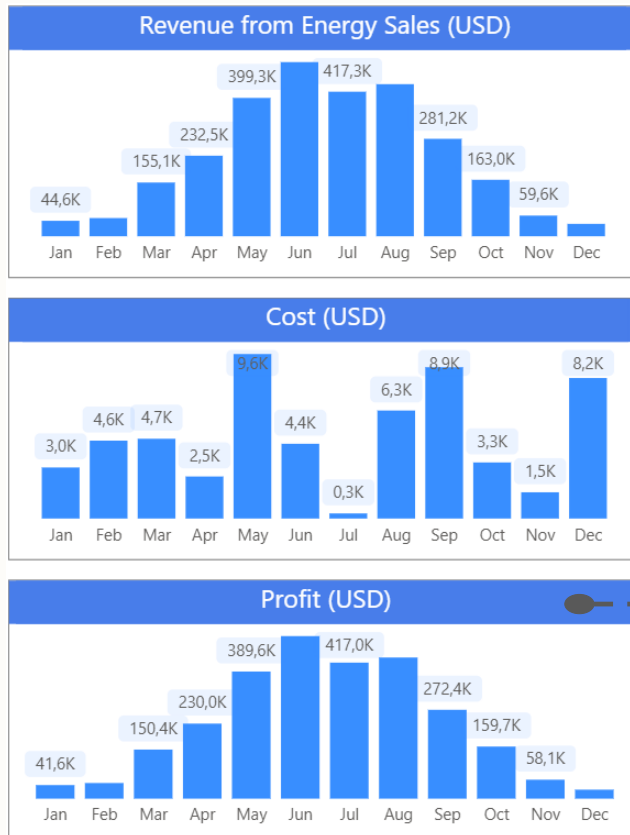
At the annual level, a **positive evolution** of the strategy is observed: corrective maintenance predominated in 2023 and 2024, while in 2025 preventive increased significantly.

This trend shows a transition toward a more planned and efficient scheme that should be sustained and deepened.





Profitability



Economic Result Calculation

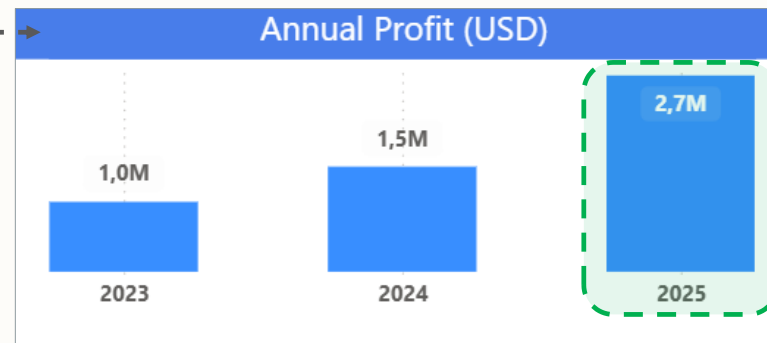
profitThe farm profit is determined as the difference between income from energy sold and associated operational costs.

$$\text{PROFIT} = \text{Revenue} - \text{Costs}$$



Result Evolution

The year-over-year comparison shows a **sustained improvement in profit**, with significant growth in 2025, driven by greater energy generation and efficient cost control, even with a larger farm.



+35%



➤ Conclusions ◀

- 1 The wind farm shows **strong profitability potential**, supported by a favorable cost structure and **opportunities for operational improvement** that could further enhance long-term performance.

Although the farm remains profitable despite certain operational inefficiencies, current performance is not fully optimized. The analysis identified avoidable efficiency losses and cost overruns which, if left unaddressed, could negatively impact future profitability and operational sustainability.



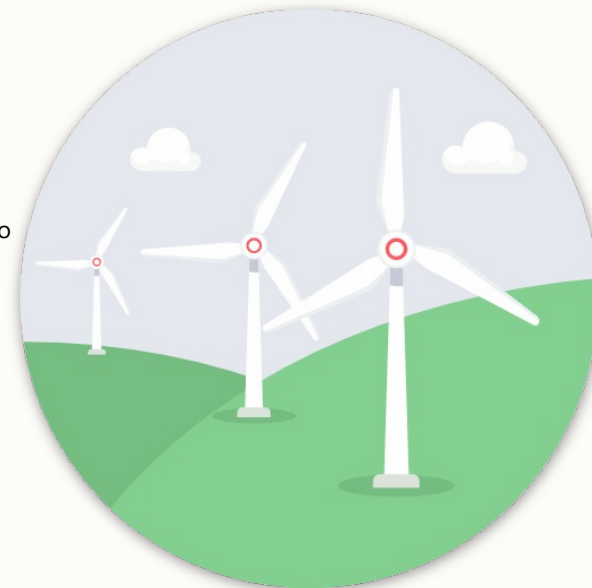
- 2 The analysis shows the need for a different asset management approach for turbines with recurring low performance, including specific technical actions and a review of the current maintenance strategy.

This performance gap reveals a clear opportunity to improve operational efficiency through technological standardization, stronger preventive maintenance practices, and prioritized intervention on critical assets, enhancing overall performance without requiring additional infrastructure expansion.



What action lines emerge from these results?

- A Lifecycle management for critical turbines (T003)**
Benchmark performance against T008, T009 and T010 to quantify the economic impact of reaching similar operating standards.
- B Strengthen preventive and predictive maintenance**
Increase scheduled inspection frequency and use historical cost and availability data to optimize maintenance planning and budget allocation.
- C Optimize maintenance scheduling based on seasonality**
Shift non-critical maintenance activities to low-generation periods to minimize generation losses.
- D Standardize future expansions with GE 2.5-116 turbines**
Prioritize GE 2.5-116 models in future expansion projects due to their superior operational performance.



Thank You!

